Hilay Trivedi

Senior Software Engineer | AI and LLM Systems

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SUMMARY

Senior Software Engineer with 5.5+ years in fully remote roles, building LLM systems since 2024. Technical lead of one of rtCamp's two AI initiative teams, reporting to the CTO. Run a hybrid retrieval system for 36 senior engineers (~300 queries/day, 83.2% recall@10) and shipped a Claude Code memory plugin to 80+ engineers that cut token usage per task by 32%. Work across Python and TypeScript with LangGraph, Weaviate, and Langfuse.

WORK EXPERIENCE

Senior Software Engineer | rtCamp (WordPress VIP Gold Agency) **Apr 2025 - Present**

Remote, India

- Technical lead of one of rtCamp's two AI initiative teams since Aug 2026, reporting to the CTO: own planning, architecture, proposals, and POC delivery, and present to leadership to drive org-wide adoption.
- Built and operate a retrieval system over **5,675 documents** from a 16-year delivery history: SQLite for entity relationships plus hybrid BM25 and BGE dense search in Weaviate, with size-adaptive chunking and overlap.
- Serves **36 senior engineers at ~300 queries/day**; **83.2% recall@10** at launch, improving through a user feedback loop.
- Added hashed, metadata-tagged content capsules and role-scoped answers for sales, architecture, and engineering users.
- Led the persistent memory layer project: shipped a Claude Code plugin with shared agent memory to **80+ engineers**, cutting token usage per task by **32%** on the company's internal ops metric.
- Designed three-tier memory scoping (org, project, personal) with git-backed markdown as source of truth and human-approved promotion; added deterministic non-LLM reranking and per-subagent model tiers for cost control. Codex adapter in progress.
- Evaluated models and agent workflows against deterministic metrics, traced in Langfuse and LiteLLM, and retired 5 workflows that did not justify their cost.
- Built AI image, video, and audio generation features for rtCamp's Digital Asset Management product.
- Tech lead for a 12-engineer team on the WordPress platform of a US prop trading firm serving 30K to 50K daily users: architecture, sprint planning, code review, and incident response; led the full platform migration.

Data Science Intern (part-time) | Tesseract Technolabs **Nov 2024 - May 2025**

Ahmedabad, India | Employer-approved, concurrent with rtCamp

- Built an end-to-end AI agent system for the hiring pipeline, from MCQ screening and a video interview stage through to the final candidate shortlist.
- Built an AI coding-interview stage that scores candidate code on unit-test pass rate and reviews it for engineering best practices.

Web Engineer | rtCamp (WordPress VIP Gold Agency) **Jan 2023 - Apr 2025**

Remote, India

- Led 8 engineers to ship 200+ landing pages in 6 weeks on a reusable PHP/ACF component system.
- Built the Gutenberg block system for Canada's largest media publisher, reaching 90+ mobile and 99 desktop Lighthouse scores on a multisite network serving millions of readers.
- Architected headless multisite networks (WPGraphQL, Next.js) that replicate Gutenberg blocks in Next.js.

Full Stack Developer | Multidots Solutions **Dec 2020 - Jan 2023**

Ahmedabad, India

- Contributed to the backend migration of a legacy .NET platform to WordPress VIP (custom plugins, WooCommerce, REST APIs), delivering a 34% post-launch sales increase.
- Built an in-house server-side rendered Gutenberg block library (PHP, React) reused across projects.

Machine Learning Intern | Llychee Inc. **Sep 2020 - Oct 2020**

Hong Kong (Remote)

- Built a LinkedIn profile scraper with ML relevance scoring against job descriptions.

SELECTED PROJECTS

Advoksha: multi-agent legal research | github.com/HILAYTRIVEDI/advoksha

- Used daily by **22 practicing lawyers** for case research and drafting while deployed; Supervisor, Researcher, Linguistic, and Drafter agents on shared Redis state (LangGraph, FastAPI, Next.js, Gemini 2.5, Supabase, Docker), grounded in High Court and Supreme Court case law.
- Chose a ~33M-parameter BGE embedding model to stay self-hostable, far below ~$250/seat/month incumbent pricing; added OCR, regional-language translation, and schema-validated contract drafting.

LessonBuild: human-in-the-loop learning agent | github.com/HILAYTRIVEDI/LessonBuild

- Turns a PDF into an approved lesson plan, objective-scoped MCQs, retryable feedback, and a personalized report (TypeScript, LangGraph.js, CopilotKit, Next.js, PostgreSQL); every interrupt is durable via a Postgres checkpointer, and an answer-leak guardrail is enforced structurally and verified with an adversarial test.

Tower of Agents: governed multi-agent platform | towerofagents.vercel.app

- Three-layer platform: Neo4j content graph with GraphRAG retrieval, LangGraph governance (agent registry, human-in-the-loop gates, audit trail), and an MCP execution layer exposing governed capabilities as an SDK.

TECHNICAL SKILLS

AI and LLM: LLM applications (Claude, Gemini, OpenAI), multi-agent orchestration (LangGraph, LangGraph.js), RAG, hybrid retrieval (BM25 + dense), GraphRAG, Weaviate, pgvector, Neo4j, BGE embeddings, structured output, MCP servers and clients, prompt-injection defense, human-in-the-loop workflows, Claude Code plugins and hooks

Evals and observability: Langfuse, LiteLLM, OpenTelemetry, recall@k, deterministic eval metrics, cost tracking, model-tier optimization

Languages: Python, TypeScript, JavaScript, PHP 8+, SQL

Backend and infra: FastAPI, NestJS, Node.js, PostgreSQL, SQLite, Redis, REST, GraphQL, WebSockets, Docker, GitHub Actions, Vercel, Supabase

Frontend: React, Next.js (App Router), CopilotKit, Tailwind CSS

OPEN SOURCE

- WordPress Core: credited contributor on releases 5.9 through 7.1 with 8 patches in Core SVN; part of the initial Core AI team and proposed the Unified AI Management Layer (LLM abstraction for plugins and themes).
- Gutenberg and Meta: Query Total color support, Embed Preview margin control, accessibility fixes, 2 Meta SVN changesets; author of the Revision Buster plugin.

EDUCATION

Executive Diploma in Machine Learning and AI, IIIT Bangalore Aug 2025 - Dec 2026 (expected)

B.Tech in Computer Engineering, L.J. Institute of Engineering and Technology (CGPA 9.6/10) 2017 - 2021